

Topic to discuss

→ Find the missing term/terms

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Some Key Concepts :

① When x values are equally spaced

$$x = 2, 4, 6, 8, 10$$

so the gap is constant, $h=2$

② Given, $y_0, y_1, y_2, \dots$

• First difference, $\Delta y_0 = y_1 - y_0$,
 $\Delta y_1 = y_2 - y_1, \dots$

• Second difference, $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$

• Third difference, $\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0 \dots$

③ If data comes from a polynomial of degree :

- degree 1 $\Rightarrow \Delta^2 = 0$
- degree 2 $\Rightarrow \Delta^3 = 0$
- degree 3 $\Rightarrow \Delta^4 = 0$

④ Interpolation Existence & uniqueness theorem,

If n distinct data points (x_i, y_i) are given,
then :

- There exists a unique polynomial $P(x)$
- of degree at most $n-1$
- that passes through all the points.

→ At most means $\leq$, not exactly.

⑤ With n data points, we can compute differences upto order $n-1$

⑥ If a function is a polynomial of degree n , then $\Delta^{n+1} = 0$

⑦ When we say, $\Delta^k = 0 \rightarrow$ It means the previous difference Δ^{k-1} is constant.

⑧ **Missing Terms Problem :**

If number of missing value = k
then choose a degree that gives at least k independent equations.

Numerical - 1

Q:- Find the missing value in the following table:

X :	2	4	6	8	10
Y :	5.6	8.6	13.9	-	35.6

Solution:- As step length, $h=2$ is constant and data is equally spaced.

We can make difference table to find missing term.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
2	5.6	3.0			
4	8.6	5.3	2.3		
6	13.9	$y_3 - 13.9$	$y_3 - 19.2$	$y_3 - 21.5$	
8	y_3	35.6 - y_3	$49.5 - 2y_3$	$68.7 - 3y_3$	$90.2 - 4y_3$
10	35.6				

Here, No. of data points = 5

So, degree of polynomial ≤ 4

If we assume degree of polynomial = 4
then $\Delta^5 = 0$, and we don't have Δ^5 .

Now assume, degree of polynomial = 3

then, $\Delta^4 = 0$, so we will get 1 equation
and we have to find 1 missing term

Now applying the property, $\Delta^4 = 0$

$$90.2 - 4y_3 = 0$$

$$y_3 = \frac{-90.2}{-4} = 22.55$$

Ans.

Numerical Problem - 2 :

Q: Evaluate the missing terms in the following table:

x	:	0	5	10	15	20	25
$f(x)$	:	6	10	?	17	?	31

Solution: Let $f(10) = a$ and $f(20) = b$

Here spacing is equal, $h=5$
So we can use the finite difference table
to find missing term.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$	$\Delta^5 f(x)$
0	6	4				
5	10	$a-10$	$a-14$	$41-3a$	$b-102+6a$	
10	a	$17-a$	$27-2a$	$b-61+3a$	$143-4b-4a$	$245-5b-10a$
15	17	$b-17$	$b-34+a$	$82-3b-a$		
20	b		$48-2b$			
25	31	$31-b$				

Here, No. of data points = 6

So, degree of polynomial ≤ 5

If we assume degree of polynomial = 5
then $\Delta^6 = 0$ and we don't have Δ^6 .

Now assume, degree of polynomial = 4
then, $\Delta^5 = 0$, so we will get 1 equation
and we have 2 missing terms to find

Now assume, degree of polynomial = 3
then, $\Delta^4 = 0$, from here we will get 2 equations
to find 2 missing terms.

So, Now applying $\Delta^4 = 0$,

$$b - 102 + 6a = 0 \Rightarrow 6a + b = 102 \text{ --- (1)}$$

$$\& \quad 143 - 4b - 4a = 0 \Rightarrow 4a + 4b = 143 \text{ --- (2)}$$

Multiply 4 in eqn --- (1)

$$24a + 4b = 408 \text{ --- (3)}$$

$$4a + 4b = 143 \text{ --- (2)}$$

Now subtract, $20a = 265$

$$\therefore a = \frac{265}{20} = 13.25$$

Now, substitute a in eqn --- (1)

$$b = 102 - 6a$$

$$= 102 - 6 \times 13.25$$

$$= 22.5$$

$$a = 13.25 \& \quad b = 22.5$$

Homework Problem :

Q: Find the missing terms in the following table :

x	:	0	1	2	3	4	5
$f(x)$	:	0	-	8	15	-	35

Solution: Let $f(1) = a$ and $f(4) = b$

Now construct the difference table.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
0	0	a	$8-2a$	$3a-9$	$b-4a-12$
1	a	$8-a$	$a-1$	$b-a-21$	$93-4b+a$
2	8	7	$b-22$	$72-3b$	
3	15	$b-15$	$50-2b$		
4	b	$35-b$			
5	35				

Since the data points = 6
then degree of polynomial ≤ 5

Assume,
degree of polynomial = 5 $\Rightarrow \Delta^6 = 0$ (Wrong assumption)

Assume,
degree of polynomial = 4 $\Rightarrow \Delta^5 = 0$ (Gives only 1 equation, wrong assumption)

So, we can assume,
degree of polynomial = 3 $\Rightarrow \Delta^4 = 0$
(Gives 2 equations).

$$\therefore b - 4a - 12 = 0 \Rightarrow 4a - b = -12 \text{ --- (1)}$$

$$93 - 4b + a = 0 \Rightarrow a - 4b = -93 \text{ --- (2)}$$

Multiply 4 in equation --- (1)

$$16a - 4b = -48 \text{ --- (3)}$$

$$a - 4b = -93 \text{ --- (2)}$$

Let subtract,

$$15a = 45 \Rightarrow a = 3$$

Now substitute, $a = 3$ in equation, --- (1)

$$\begin{aligned} 1 \quad 4a - b &= -12 \Rightarrow b = 12 + 4a \\ &= 12 + 4 \times 3 \\ &= 24 \end{aligned}$$

Hence, $a = 3$ and $b = 24$

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